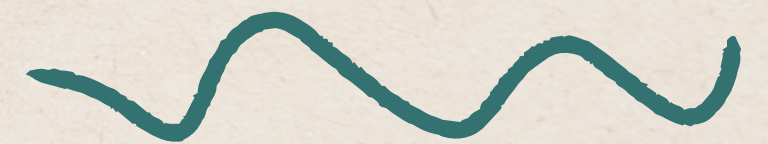




SkyBook



Airport Configuration

Purpose

Configure airport entity properties and validation rules.

Configurations

Name

Required.

Maximum length: 100 characters.

Reason: Airport name is essential information and should not be empty.

Code

Required.

Maximum length: 10 characters.

Unique.

Reason: Every airport must have a unique code such as CAI or DXB.

City

Required.

Maximum length: 100 characters.

Reason: Used to identify the airport location.

Country

Required.

Maximum length: 100 characters.

Reason: Used to identify the airport country.

Aircraft Configuration

Purpose

Configure aircraft information and capacity.

Configurations

Name

Required.

Maximum length: 100 characters.

Reason: Each aircraft must have a unique identifiable name.

Capacity

Required.

Must be greater than zero.

Reason: An aircraft cannot have zero seats.

Seat Configuration

Purpose

Configure aircraft seats.

Configurations

SeatNumber

Required.

Maximum length: 10 characters.

Reason: Stores seat numbers such as A1, B5, C10.

Class

Enum.

Possible values:

Economy

Business

FirstClass

Aircraft Relationship

One Aircraft has many Seats.

Every Seat belongs to one Aircraft.

Delete Behavior

Restrict

Reason Prevent deleting an aircraft that contains seats.

Flight Configuration

Purpose

Configure flight information and flight relationships.

Configurations

FlightNumber

Required.

Maximum length: 20 characters.

Unique.

Reason Each flight must have a unique identifier.

Price

Required.

Decimal(18,2).

Reason Stores ticket prices accurately.

Status

Enum.

Stored as string.

Possible values:

Scheduled

Delayed

Cancelled

Completed

Reason Ensures valid flight statuses.

Departure Airport Relationship

One Airport can be used by many Flights.

Every Flight has one Departure Airport.

Delete Behavior

Restrict

Reason An airport used by flights cannot be deleted.

Arrival Airport Relationship

One Airport can be used by many Flights.

Every Flight has one Arrival Airport.

Delete Behavior

Restrict

Aircraft Relationship

One Aircraft can perform many Flights.

Every Flight uses one Aircraft.

Delete Behavior

Restrict

Booking Configuration

Purpose

Configure reservation information.

Configurations

BookingReference

Required.

Maximum length: 50 characters.

Unique.

Reason Every booking must have a unique reference number.

BookingDate

Required.

Reason Stores reservation creation date.

TotalPrice

Required.

Reason Preserves the booking price even if flight prices change later.

Status

Enum.

Stored as string.

Possible values:

Pending

Confirmed

Cancelled

User Relationship

One User can have many Bookings.

Every Booking belongs to one User.

Delete Behavior

Cascade

Reason If the user is removed, related bookings can be removed automatically.

Flight Relationship

One Flight can have many Bookings.

Every Booking belongs to one Flight.

Delete Behavior

Restrict

Reason Flights with bookings should not be deleted accidentally.

Seat Relationship

One Seat can have many Bookings over time.

Every Booking references one Seat.

Delete Behavior

Restrict

ApplicationUser Configuration

Purpose: Store customer account information and support authentication and booking management.

Properties Configuration:

Property

FullName

Required, Max Length 100

Country

Max Length 100

Email

Unique

PhoneNumber

Max Length 20

Relationships:

ApplicationUser → Bookings

One-to-Many

Delete Behavior:

Restrict

DbContext

AppDbContext Responsibilities
Manage Database Tables

- ApplicationUser
- Airport
- Aircraft
- Seat
- Flight
- Booking

Apply Configurations

- AirportConfiguration
- AircraftConfiguration
- SeatConfiguration
- FlightConfiguration
- BookingConfiguration
- ApplicationUserConfiguration

Database Connection

Connection String

ViewModel

ViewModels are used to transfer only the required data between Controllers and Views. They improve security, support validation, combine data from multiple entities, and prevent exposing database entities directly to the presentation layer.

ViewModel

RegisterViewModel

Properties

FullName

Email

Password

ConfirmPassword

Country

PhoneNumber

Validation

FullName Required

Email Required

Valid Email

Password Required

Minimum Length 6

ConfirmPassword must match Password

Used In

Register Page

LoginViewModel

Properties

Email

Password

RememberMe

Validation

Email Required

Password Required

Used In

Login Page

FlightSearchViewModel.

Properties

DepartureAirportId

ArrivalAirportId

TravelDate

PassengerCount

Validation

Departure Airport Required

Arrival Airport Required

Travel Date Required

Departure ≠ Arrival

Used In

Search Flights Page

FlightResultViewModel

Properties

FlightId

FlightNumber

DepartureAirport

ArrivalAirport

DepartureTime

ArrivalTime

Price

AvailableSeats

Status

Used In

Search Results Page

FlightDetailsViewModel

Properties

FlightId

FlightNumber

DepartureAirport

ArrivalAirport

AircraftName

DepartureTime

ArrivalTime

Price

Status

AvailableSeats

Used In

Flight Details Page

FlightCreateViewModel

Properties

FlightNumber

DepartureAirportId

ArrivalAirportId

AircraftId

Price

DepartureTime

ArrivalTime

Status

Validation

Required Fields

Price > 0

Arrival Time > Departure Time

Departure Airport ≠ Arrival Airport

Used In

Create Flight Page

Edit Flight Page

AirportViewModel

Properties

Name

Code

City

Country

Validation

All Required

Code Unique

Used In

Create Airport

Edit Airport

AircraftViewModel

Properties

Name

Capacity

Validation

Name Required

Capacity > 0

Used In

Create Aircraft

Edit Aircraft

SeatSelectionViewModel

Properties

FlightId

SeatId

SeatNumber

SeatClass

IsAvailable

Validation

Seat must be available

Used In

Seat Selection Page

BookingViewModel

Properties

FlightId

SeatId

PassengerName

PassengerEmail

PassengerPhone

Validation

Required Fields

Seat Available

Used In

Booking Page

BookingDetailsViewModel

Properties

BookingReference

CustomerName

FlightNumber

DepartureAirport

ArrivalAirport

SeatNumber

BookingDate

TotalPrice

Status

Used In

Booking Details Page

My Bookings Page

DashboardViewModel

Properties

TotalFlights

TotalAirports

TotalAircrafts

TotalCustomers

TotalBookings

RecentBookings

RecentFlights

Used In

Admin Dashboard

Thank you